XRPL Validator — Daily Engineering Report · 2026-04-16

Period: 2026-04-16 00:00 to 24:00 US Eastern (2026-04-16 04:00 to 2026-04-17 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 7 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-04-16): 0.94565 (1,064 missed of 19,576). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(sync): skip retry+full-scan on mismatch, just log and move on** (`a779791`, xrpl-validator) The 3x retry + 5min full scan never healed a real divergence - it only helped with transient rippled RPC staleness. In practice it blocked sync for 5+ minutes after every mismatch, preventing any further mismatches from being captured for diagnosis.
- **fix(sync): update hasher during fast-path gap catchup** (`5957491`, xrpl-validator) ROOT CAUSE FOUND: When ws_sync falls behind (network closed multiple ledgers since our last sync), it processes gap ledgers via "fast path" which writes to state.rocks but skips the hash check. The fast path updated leaf_cache (used only for disk persistence) but did NOT update the live hasher.
- **perf(sync): hash-check every ledger + add per-stage timing** (`4f2122c`, xrpl-validator) Fix #1: force compute_hash=true on every ledger. Previously only the last ledger in a gap-catchup batch got hash-verified, so the validation match count was capped by the "catch up to tip" rate. Now every ledger applied is a ledger verified.
- **fix(sync): fetch account_hash for every ledger, not just batch tail** (`6cc3fb6`, xrpl-validator) compute_hash was always true after the last fix, but account_hash was still only populated when process_seq == closed_seq. Gap ledgers computed their hash and had nothing to compare against — so no MATCH/MISMATCH log fired and shadow hash stayed at 0.

Docs and operations

- **diag(sync): always snapshot hasher so overlay_diff dumps every mismatch** (`e80aaad`, xrpl-validator)

- **diag: add check_key tool to look up SLE in state.rocks read-only** (`49553ad`, xrpl-validator)
- **diag(ffi): log sender AccountRoot sequence at terPRE_SEQ + tesSUCCESS** (`9711baf`, xrpl-validator)

Independence program

No independence record for UTC day 2026-04-16; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
04-16 10:25	—	xrpl-validator	`a779791`	fix(sync): skip retry+full-scan on mismatch, just log and move on
04-16 11:04	—	xrpl-validator	`e80aaad`	diag(sync): always snapshot hasher so overlay_diff dumps every mismatch
04-16 12:26	—	xrpl-validator	`49553ad`	diag: add check_key tool to look up SLE in state.rocks read-only
04-16 13:13	—	xrpl-validator	`9711baf`	diag(ffi): log sender AccountRoot sequence at terPRE_SEQ + tesSUCCESS
04-16 15:17	—	xrpl-validator	`5957491`	fix(sync): update hasher during fast-path gap catchup
04-17 00:41	—	xrpl-validator	`4f2122c`	perf(sync): hash-check every ledger + add per-stage timing
04-17 00:59	—	xrpl-validator	`6cc3fb6`	fix(sync): fetch account_hash for every ledger, not just batch tail

Times in UTC. A commit appears on the day it was committed to main or merged into it.